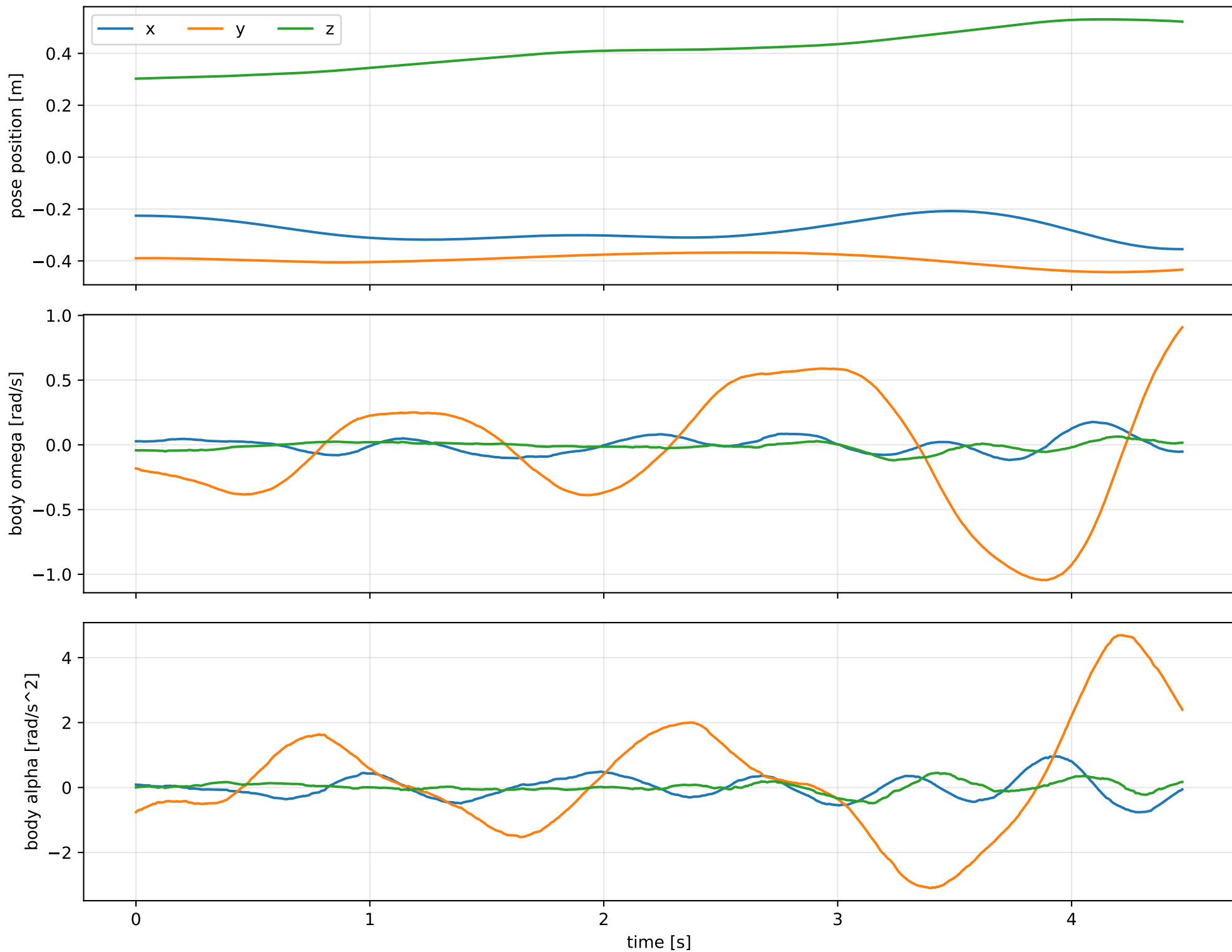
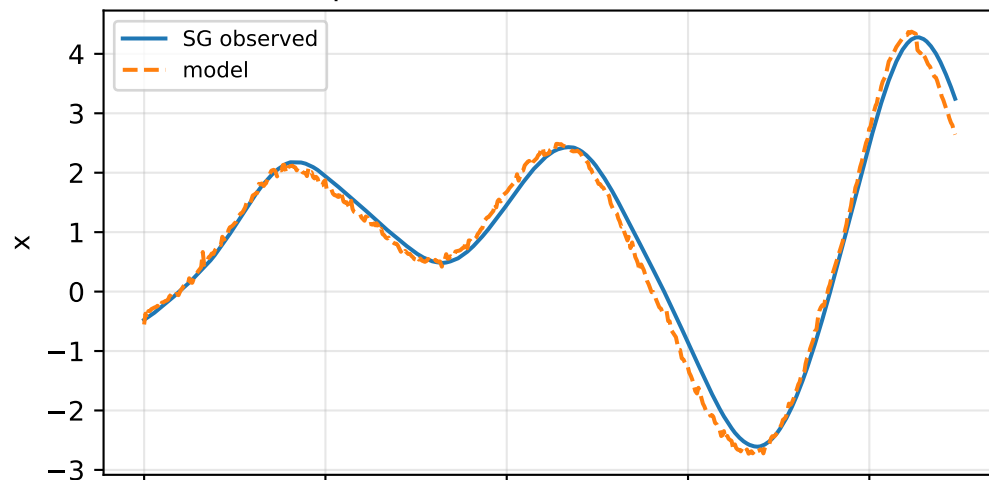


naive_all: SG trajectory and derived motion

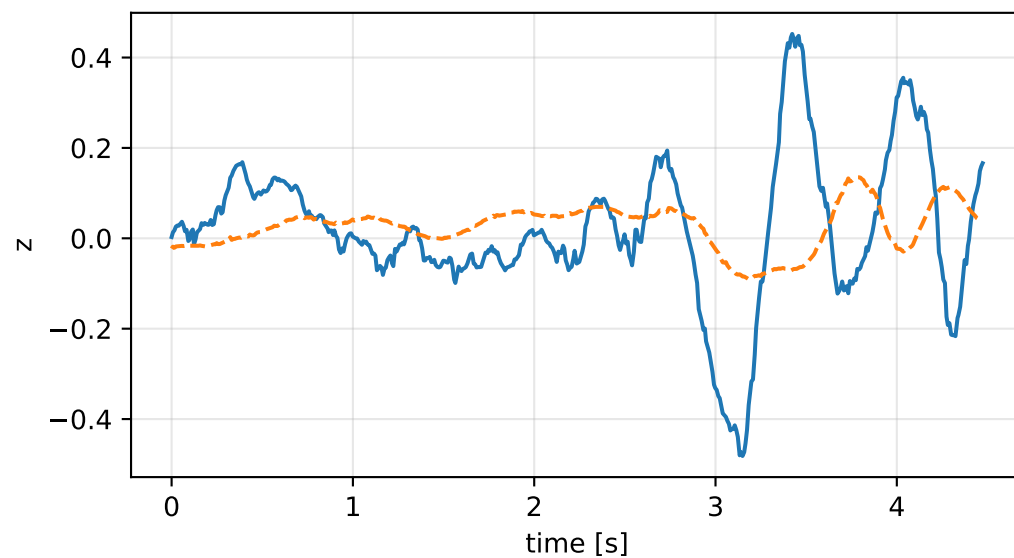
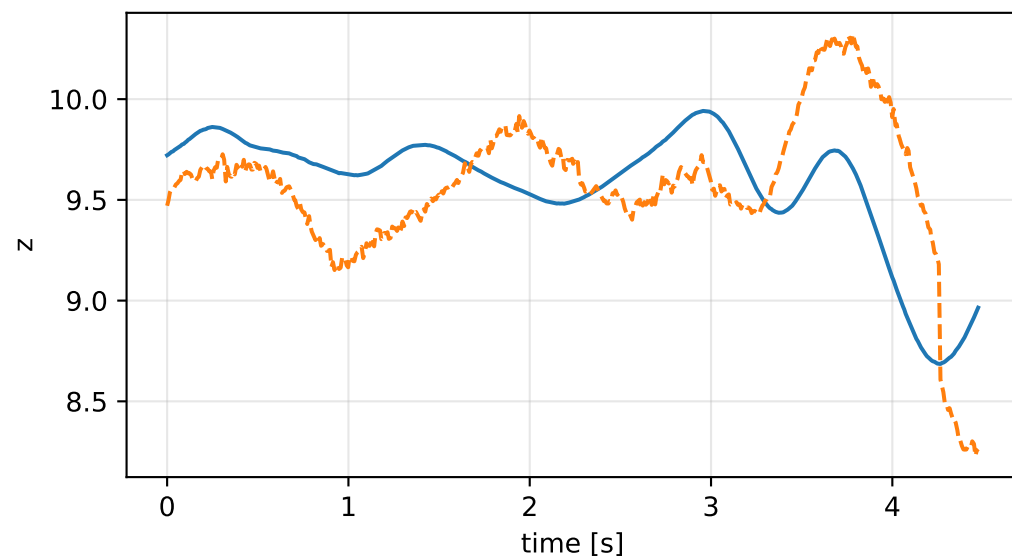
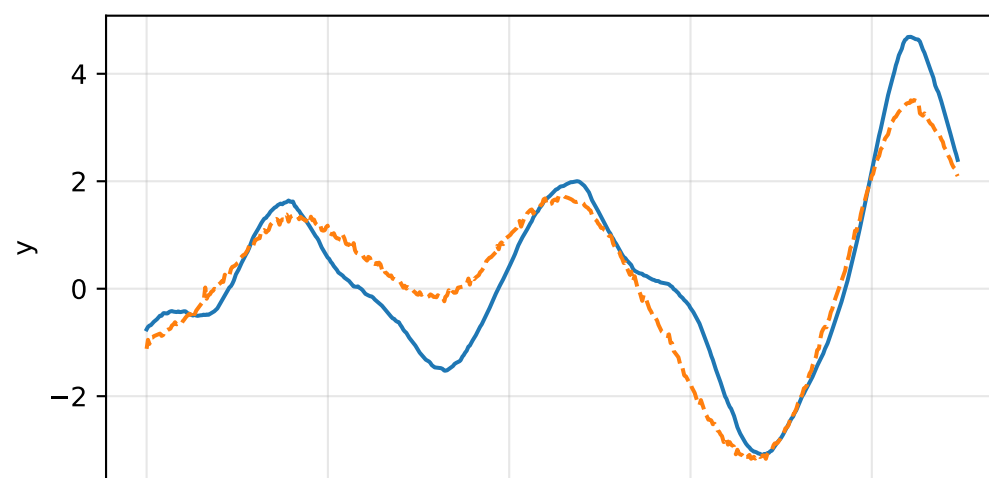
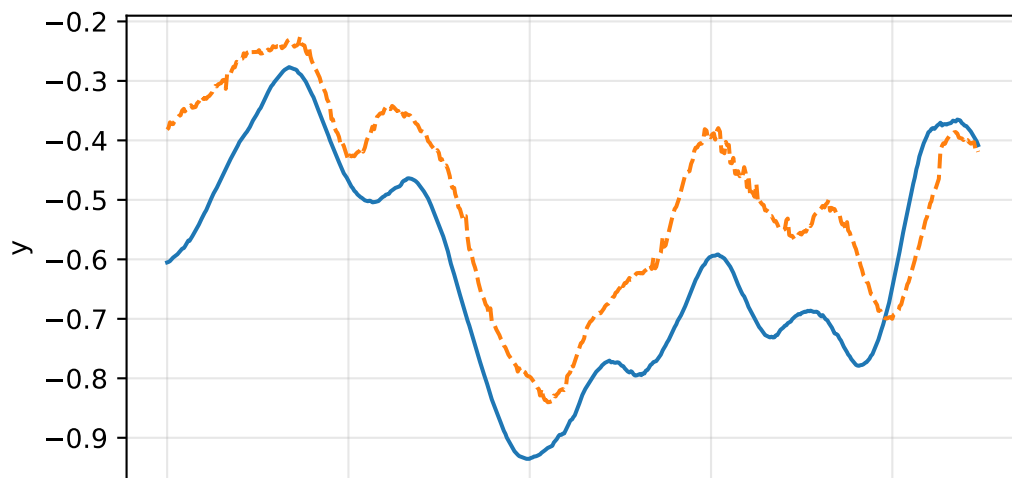
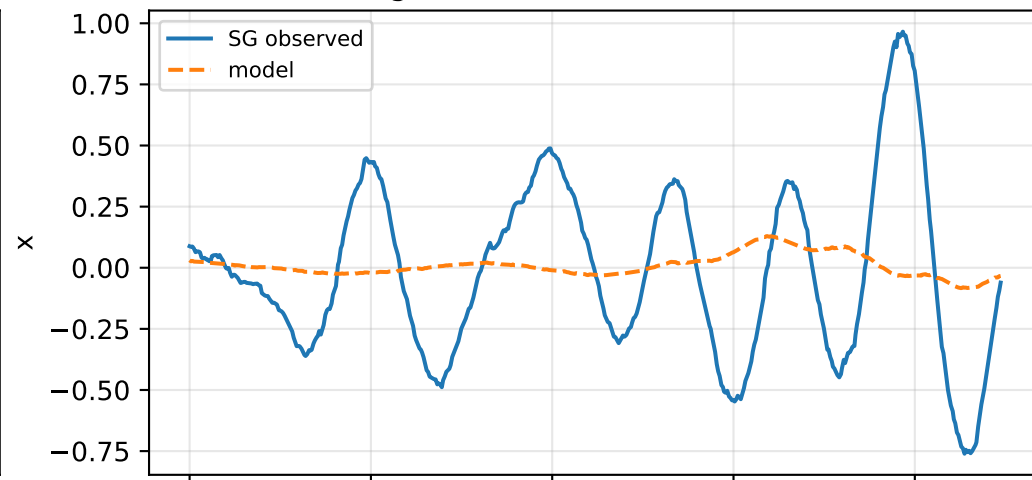


naive_all: acceleration residual objective

specific acceleration [m/s^2]

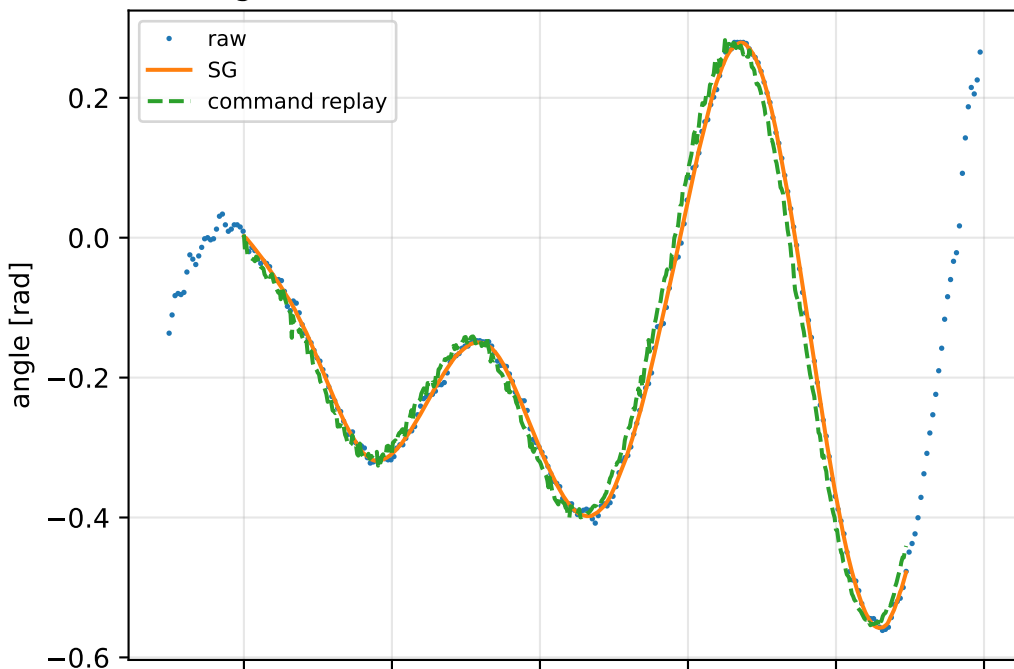


angular acceleration [rad/s^2]

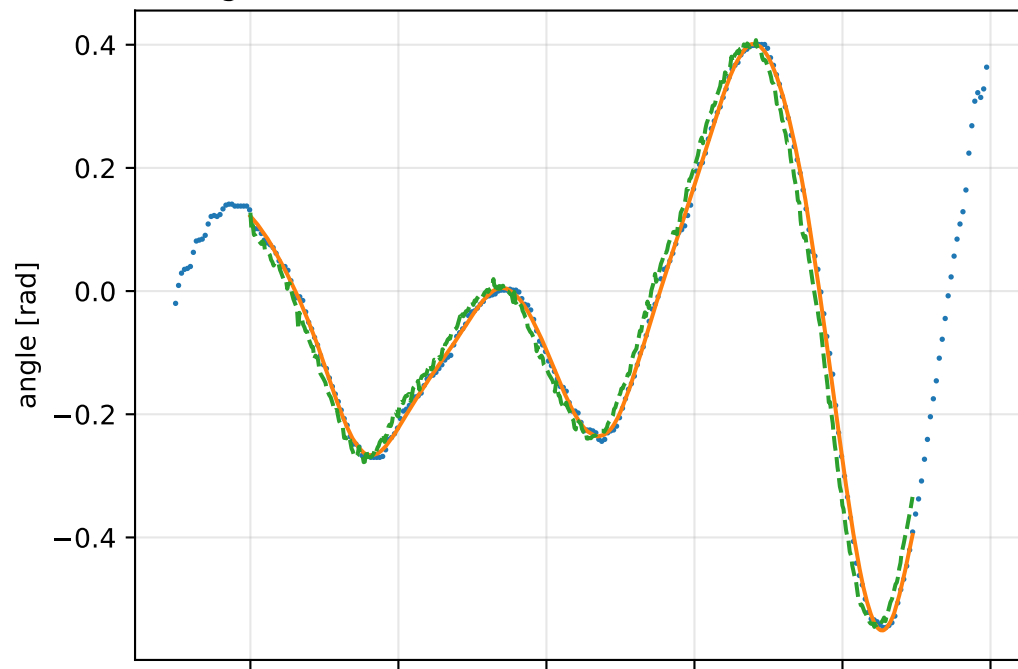


naive_all: gimbal measurement audit (objective=command_replay)

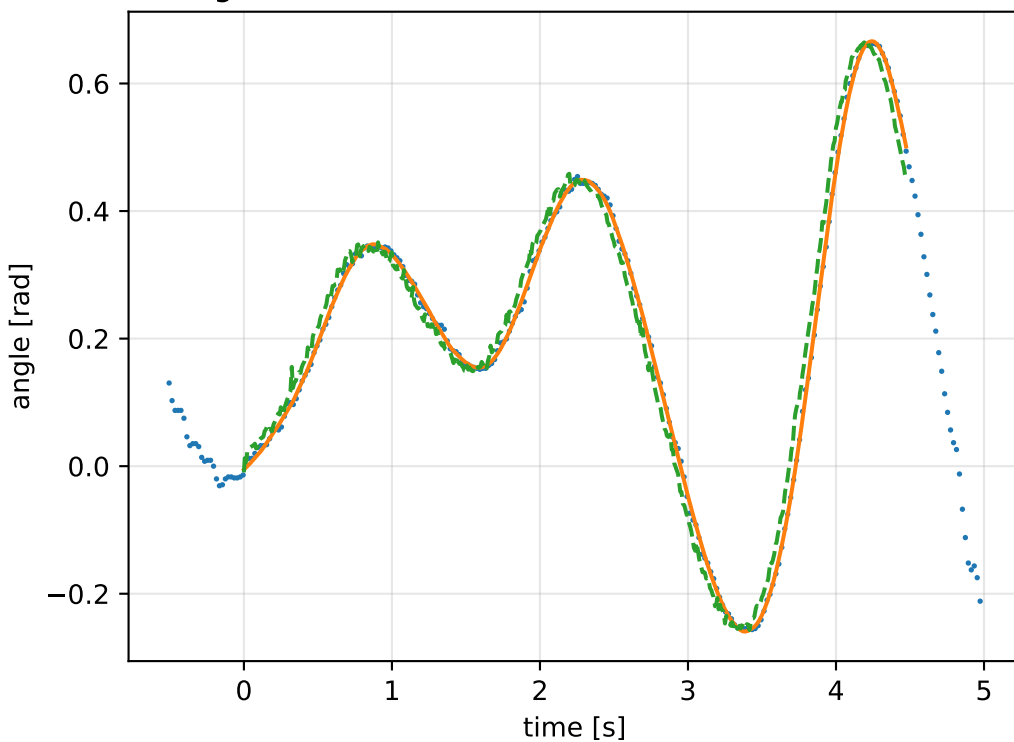
gimbal 1: RMSE=0.027, max=0.07529 rad



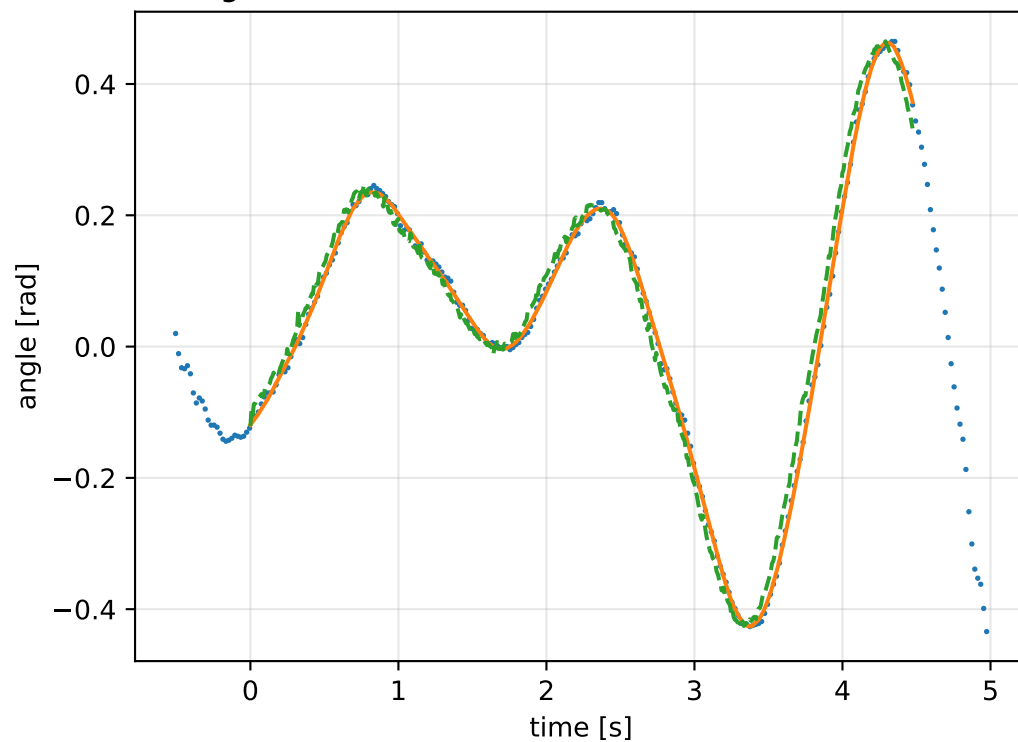
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

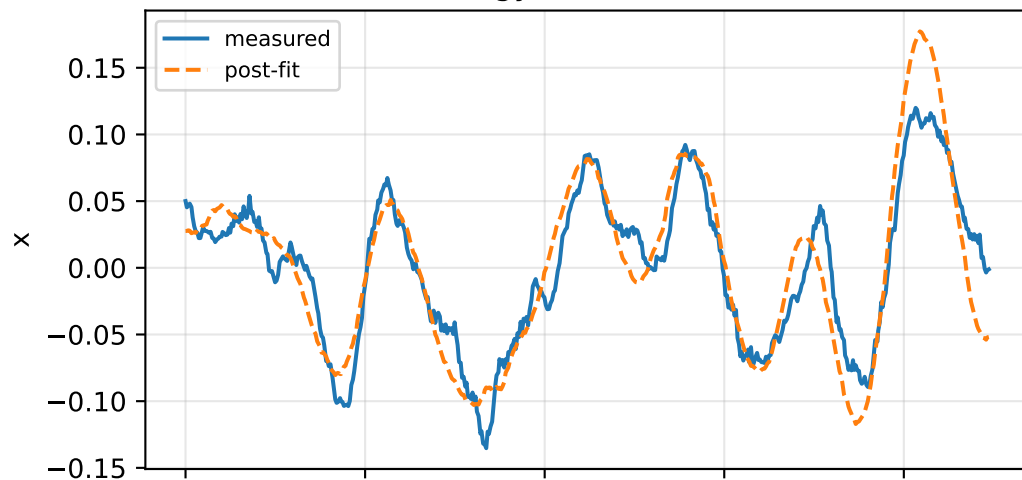


gimbal 4: RMSE=0.02598, max=0.07261 rad

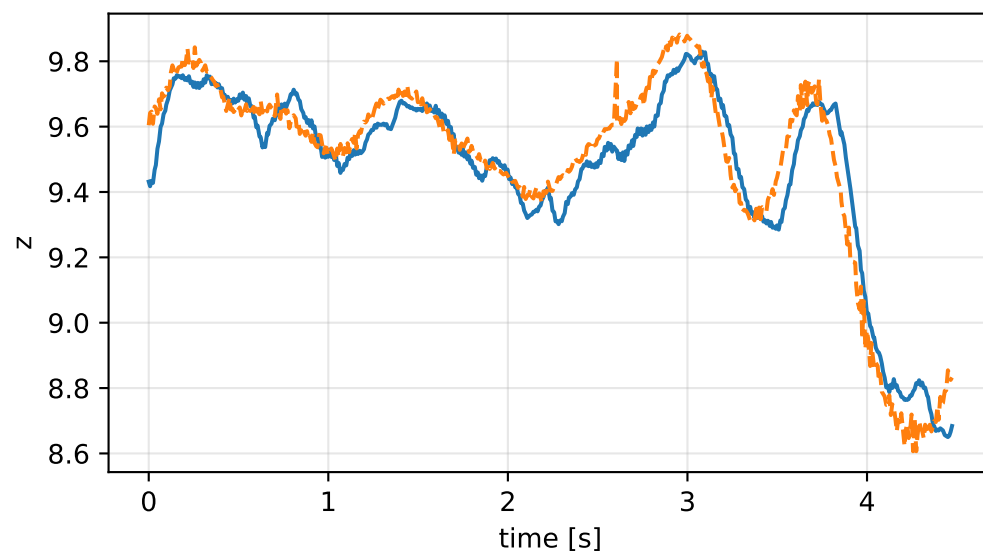
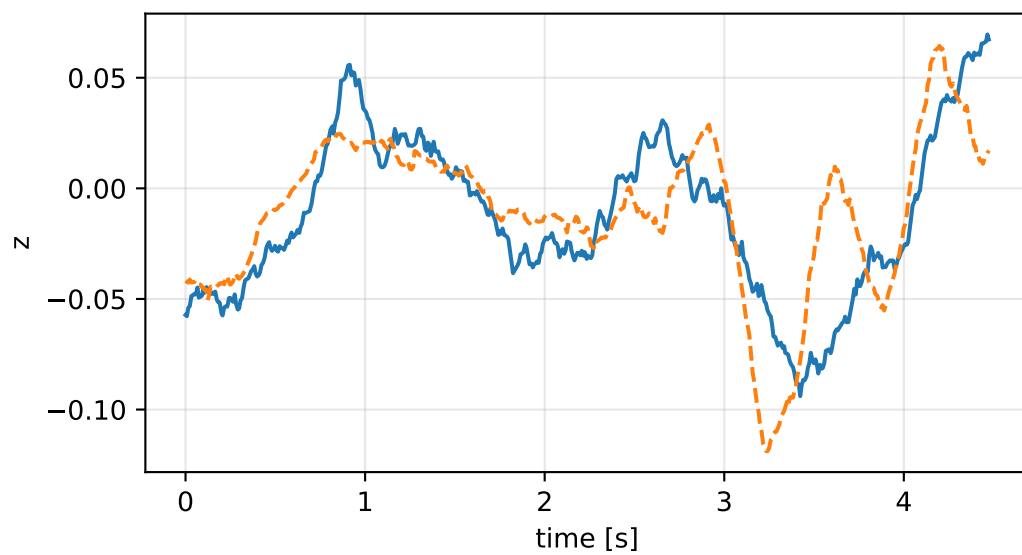
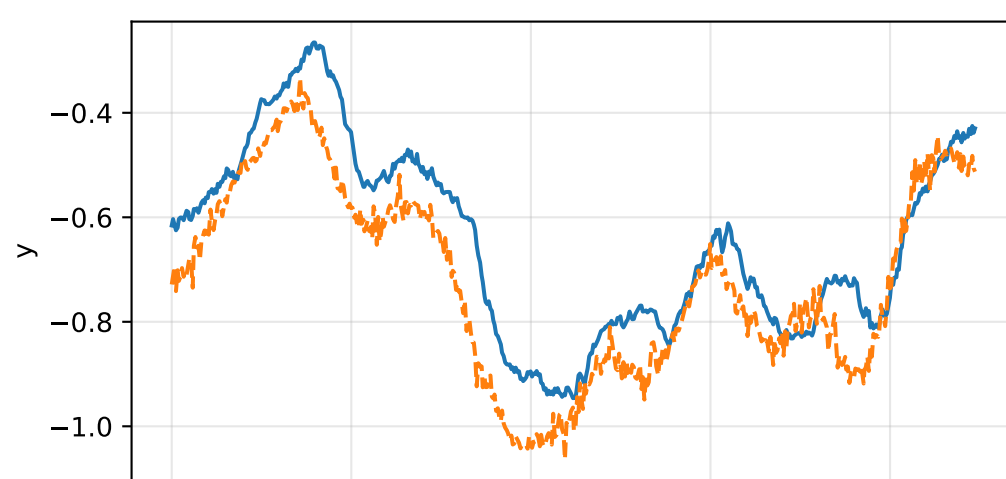
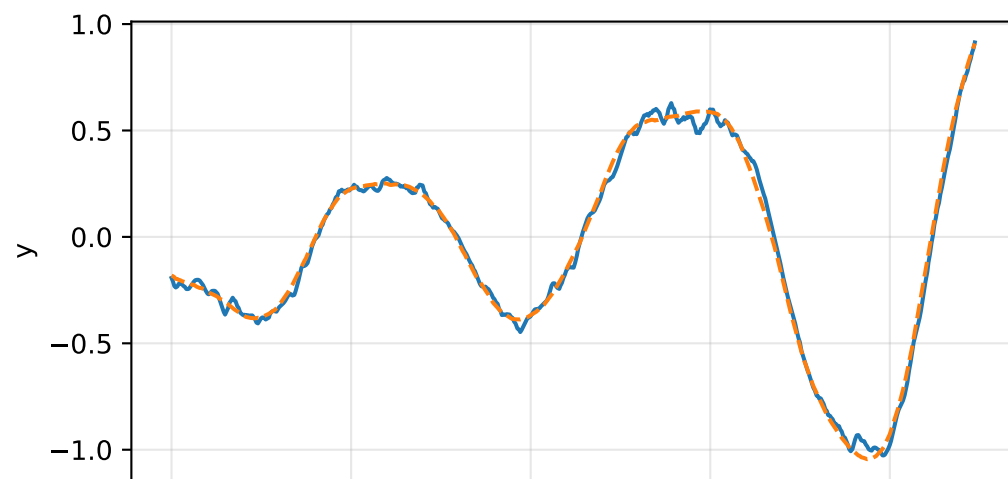
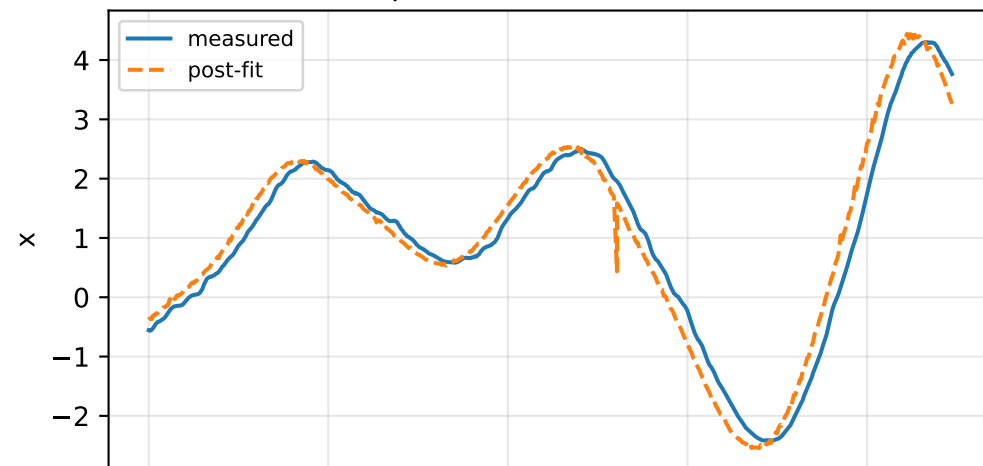


naive_all: IMU comparison (diagnostic only)

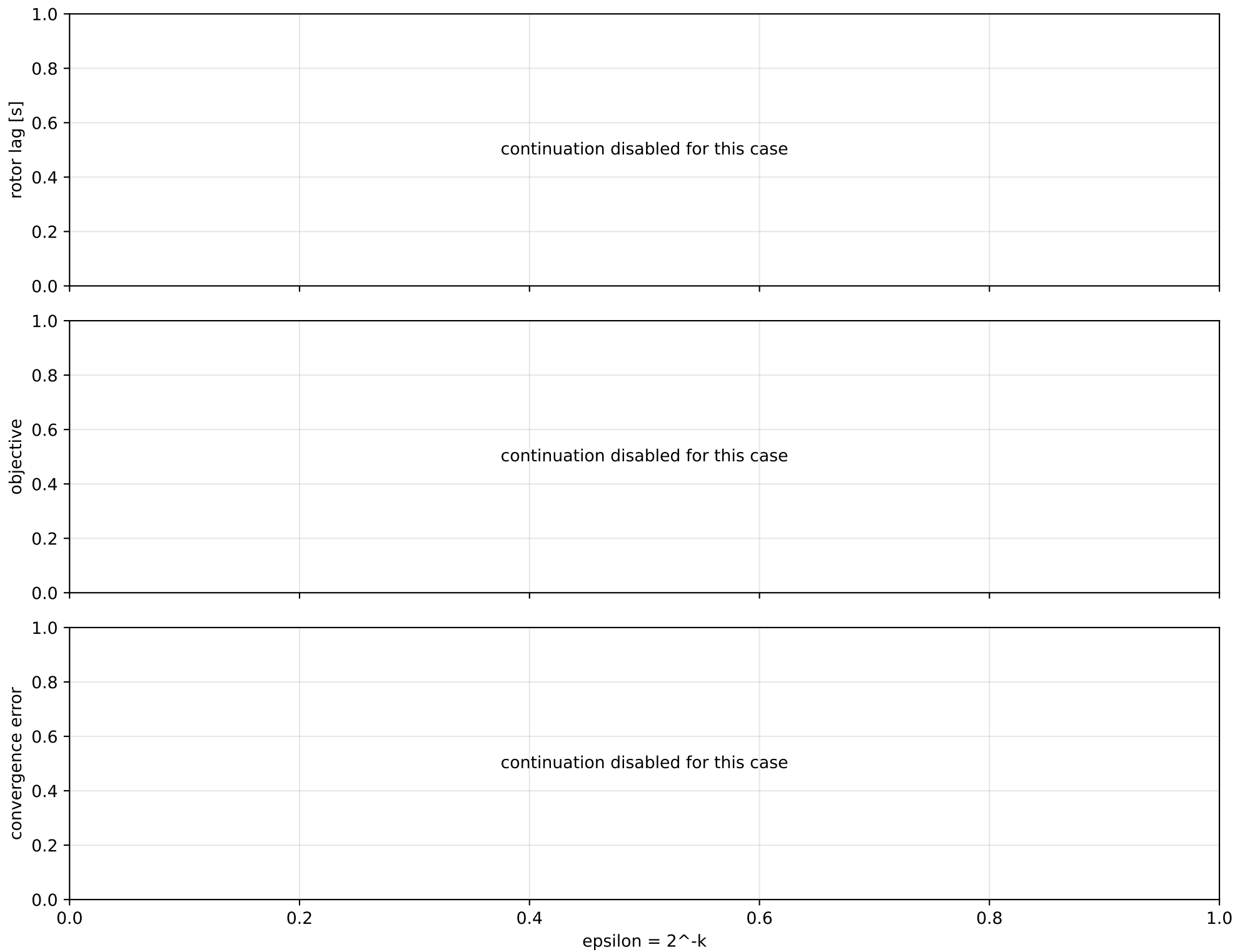
gyro [rad/s]



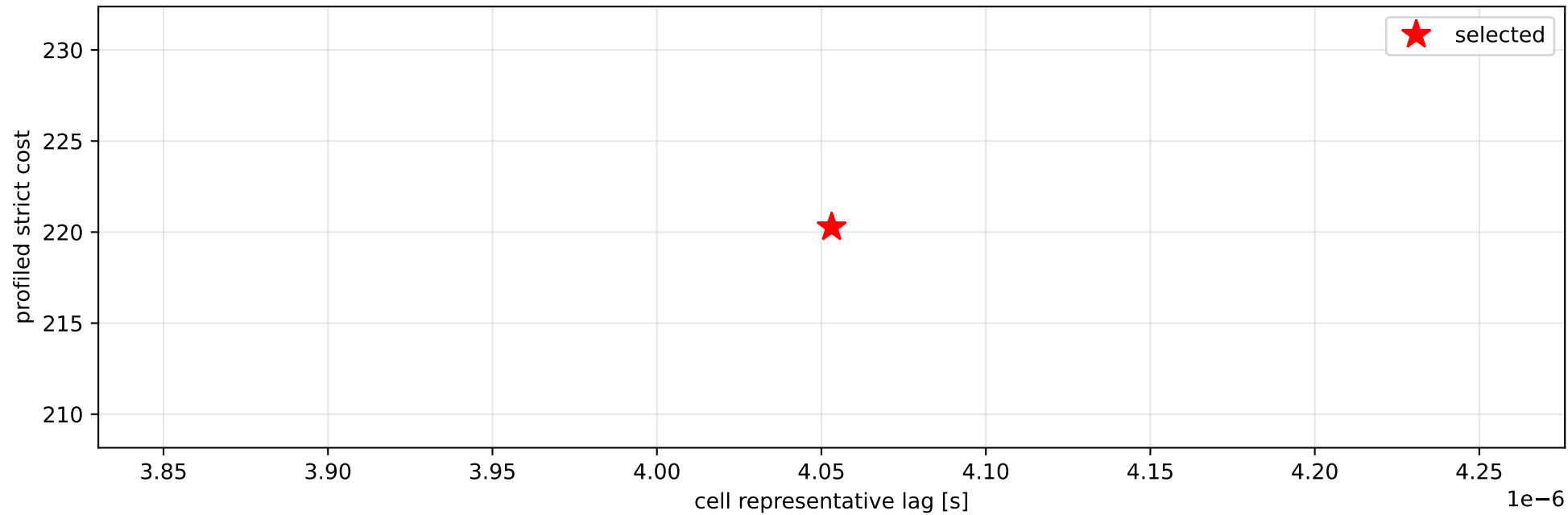
specific force [m/s^2]



naive_all: smooth-to-strict continuation



naive_all: exact strict-ZOH lag cells



selected (0, 8.10623169e-06] s; final smooth 0 s; data boundary=False

naive_all: parameter estimate

Case: naive_all

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 126.928106

inertia [kg m²]:

[[9.69139527e+02 2.50167339e+01 1.32452254e-01]

[2.50167339e+01 7.45723898e+00 -2.62791829e+01]

[1.32452254e-01 -2.62791829e+01 9.75071613e+02]]

CoG body [m]: [-0.22670478 -0.00844707 -0.0702305]

force effectiveness: [9.03154733e+01 3.37030861e-06 3.85927842e-06 8.65064528e+01]

scale-free J/m [m²]:

[[7.63534222e+00 1.97093730e-01 1.04352187e-03]

[1.97093730e-01 5.87516762e-02 -2.07039904e-01]

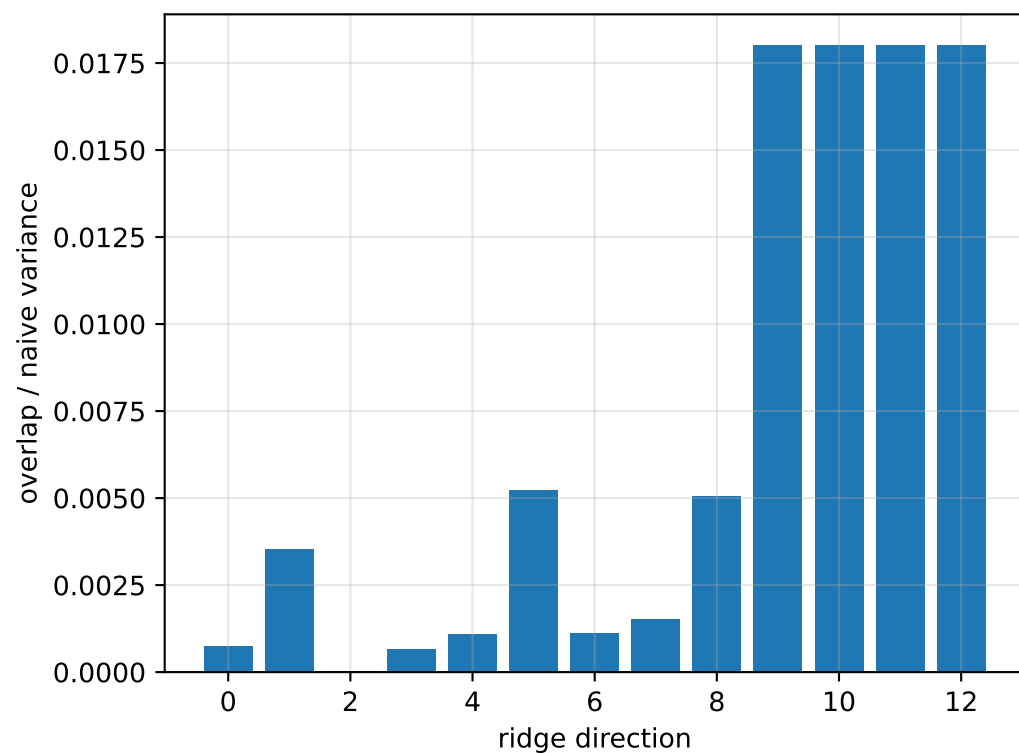
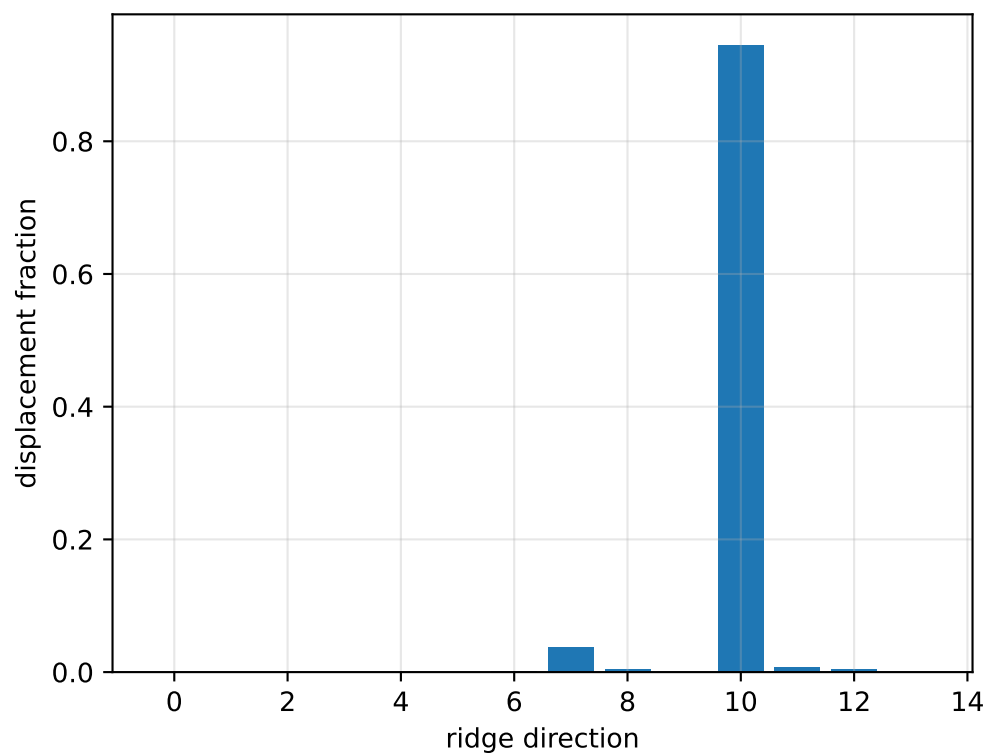
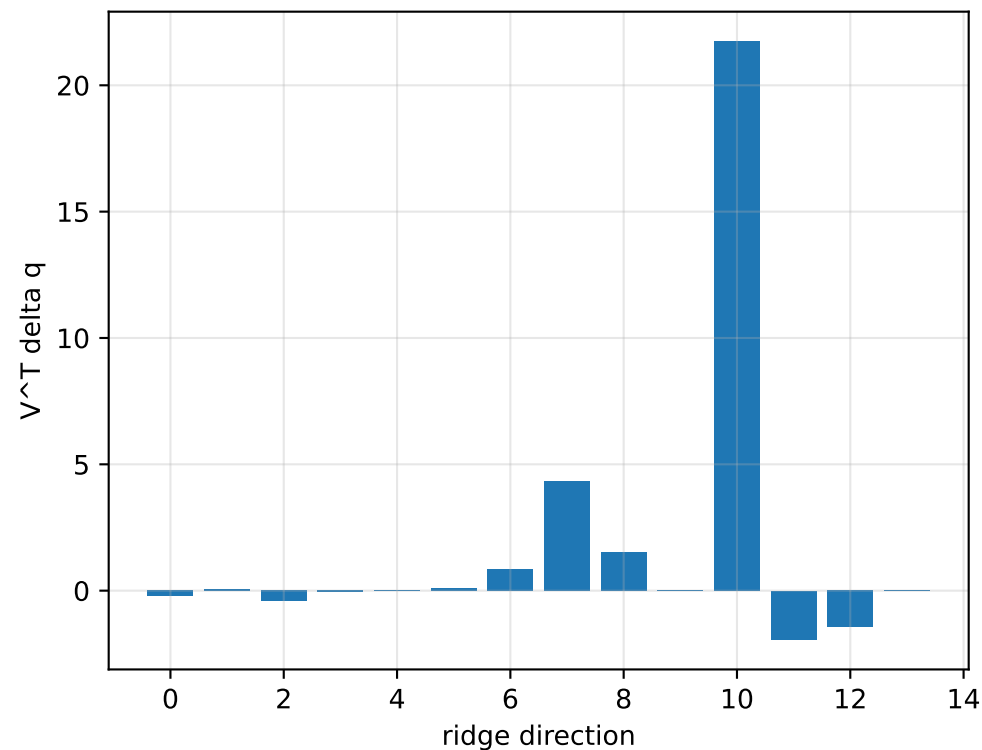
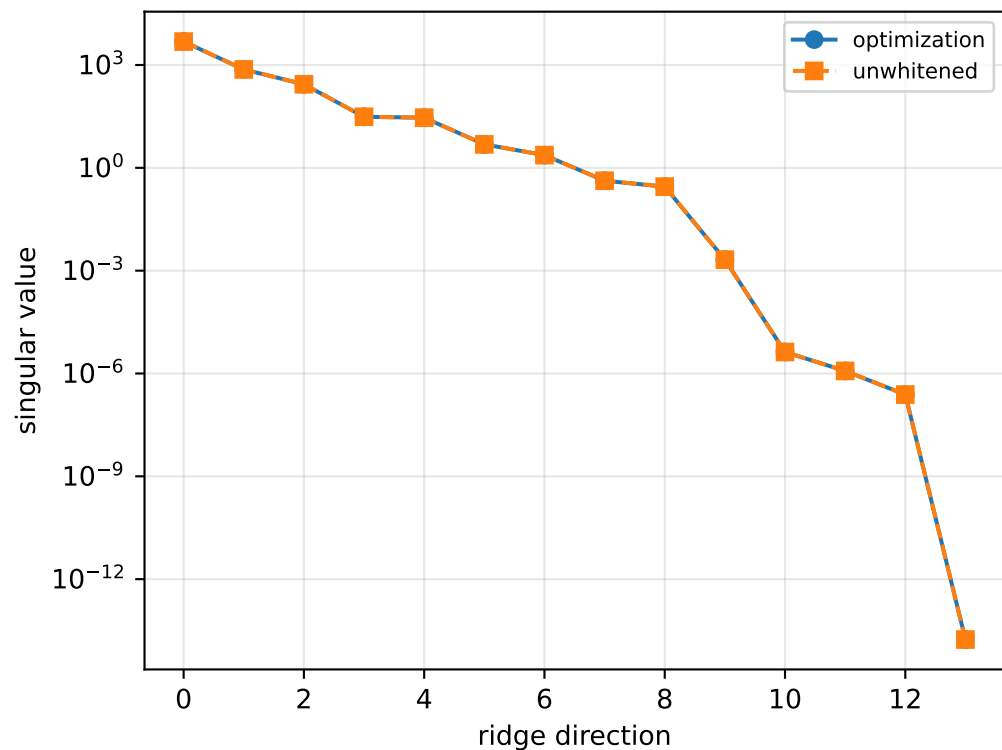
[1.04352187e-03 -2.07039904e-01 7.68207802e+00]]

scale-free f/m: [7.11548262e-01 2.65528945e-08 3.04052312e-08 6.81538987e-01]

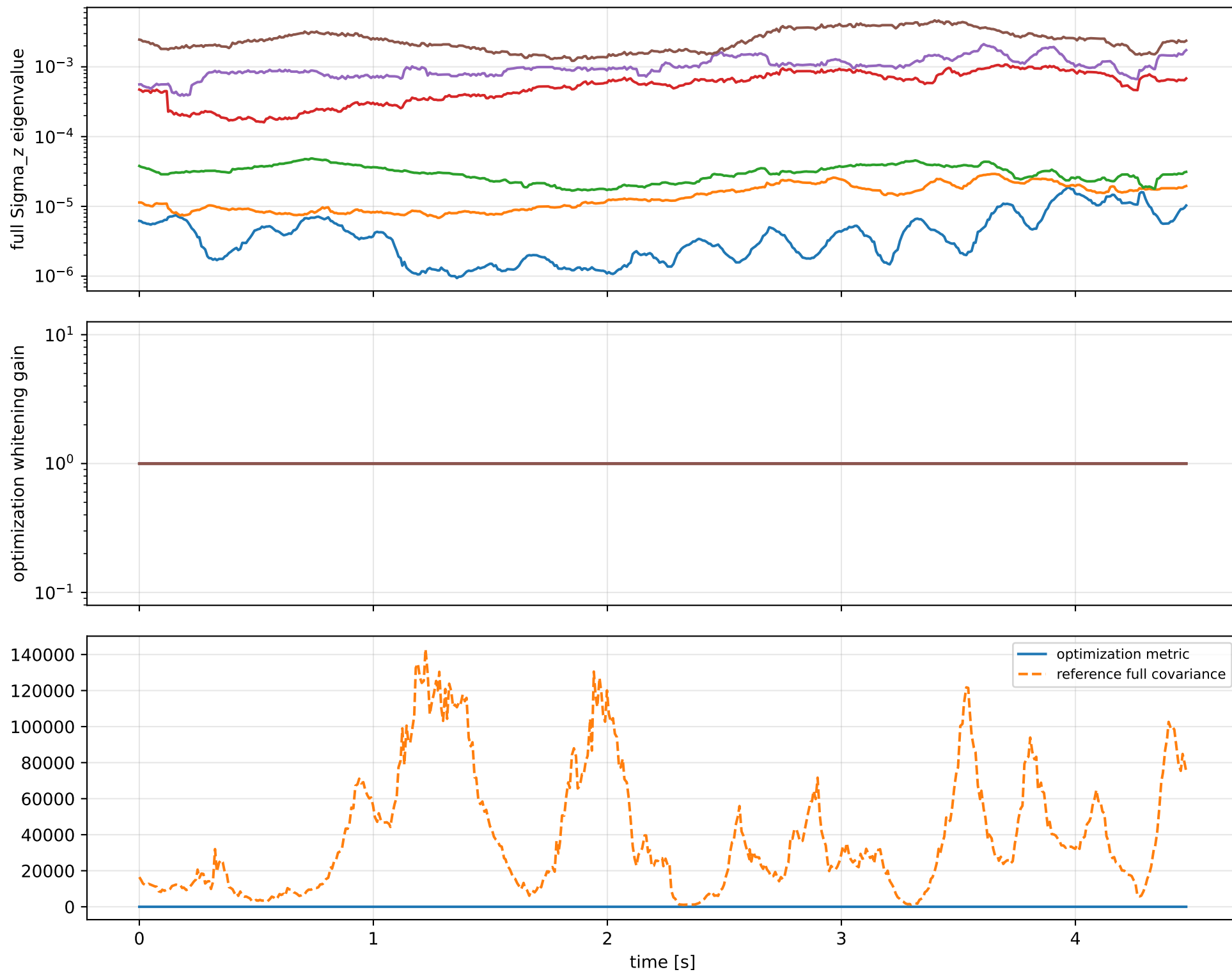
rotor lag cell: (0, 8.10623169e-06] s

representative: 0 s

naive_all: ridge spectrum (rank 13, nullity 1, ||jv||=2.29e-13)



naive_all: optimization vs reference covariance



naive_all: wrench / closure (force max 3.91e-13, torque max 2.27e-13)

